

BREST CANCER CLASSIFICATION

1. Problem Statement

The objective of this project is to develop a machine learning system that classifies breast tumors as benign (non-cancerous) or malignant (cancerous). The project aims to train and compare classification models and evaluate their performance using appropriate evaluation metrics.

2. Dataset Used

The **Breast Cancer Wisconsin (Diagnostic)** dataset was obtained from Kaggle. It contains 569 records and 30 numerical features describing cell nuclei, including radius, texture, perimeter, area, and smoothness. The target variable, diagnosis, indicates whether a tumor is benign or malignant.

	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothness_mean	compactness_mean
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280

5 rows × 33 columns

The dataset was cleaned by checking missing values and duplicates and removing unnecessary columns. Exploratory data analysis was performed using visualizations to understand the data distribution and relationships between features.

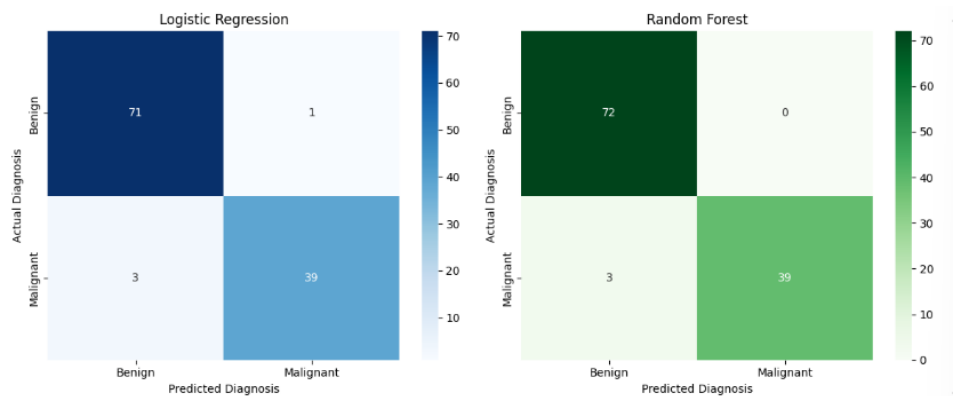
3. Models Selected

Two supervised machine learning algorithms were implemented:

Logistic Regression and **Random Forest Classifier**. Logistic Regression predicts the probability of a tumor belonging to a particular class, while Random Forest combines multiple decision trees to make predictions. The dataset was divided into 80% training data and 20% testing data. Feature scaling was applied for Logistic Regression.

4. Results and Best-Performing Model

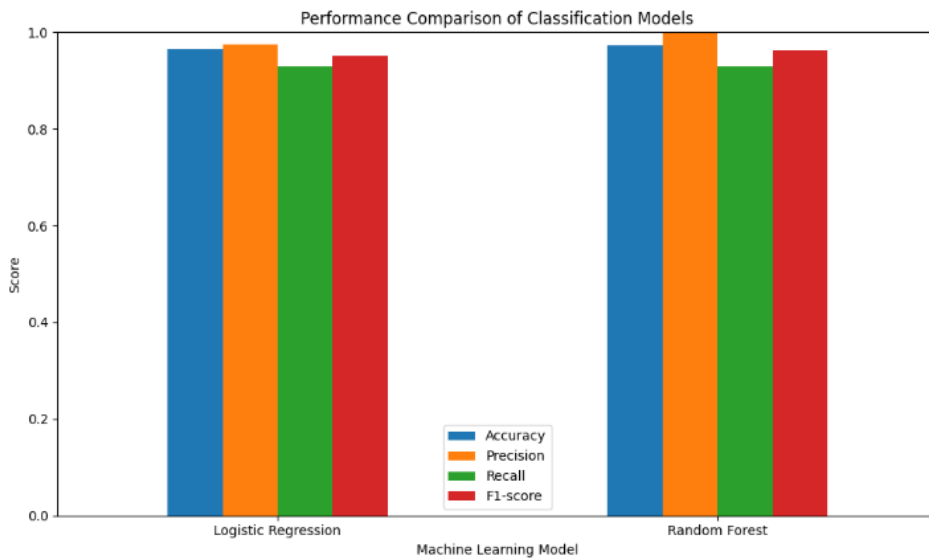
Both models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. A comparison table and bar chart were used to analyze their performance.



The best-performing model was identified based on the actual evaluation results, considering the F1-score and recall for malignant cases. These metrics are important because failing to identify a malignant tumor can have serious consequences.

Model Comparison:

	Model	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	0.9649	0.975	0.9286	0.9512
1	Random Forest	0.9737	1.000	0.9286	0.9630



5. Limitations

The project uses a relatively small dataset, so its results may not generalize to all patient populations. Model performance also depends on the quality and representativeness of the data. High accuracy alone does not guarantee reliable medical predictions.

6. Future Improvements

Future work could include testing additional algorithms, applying cross-validation, performing hyperparameter tuning, and evaluating the models on independent datasets.

Feature selection and class-specific performance analysis could also improve the evaluation process. Further clinical validation would be necessary before any real-world medical application.